

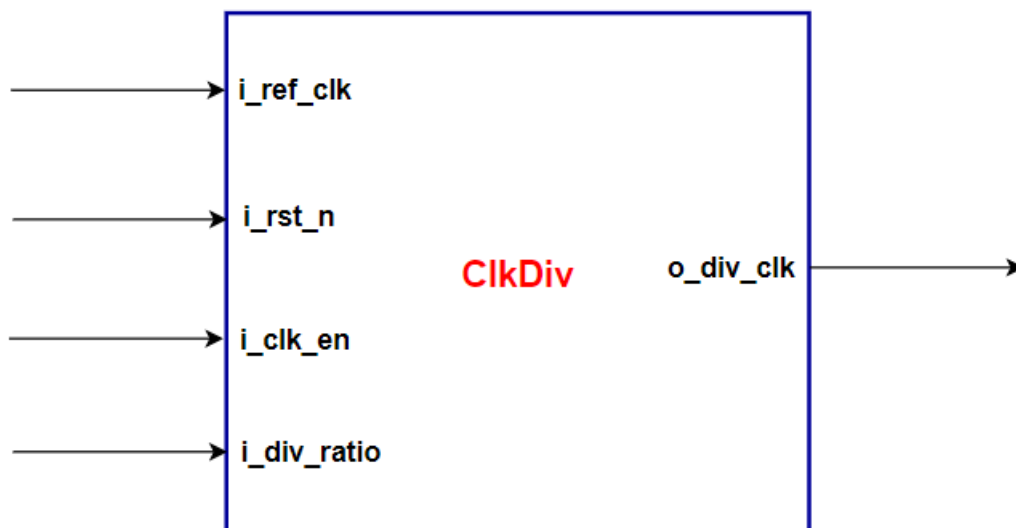
Integer Clock Divider

Introduction: -

A clock divider is a circuit that takes an input signal of a frequency **fin** and generates an output signal of a frequency **fout**, where

fout = **fin** / **n** and "n" is an integer

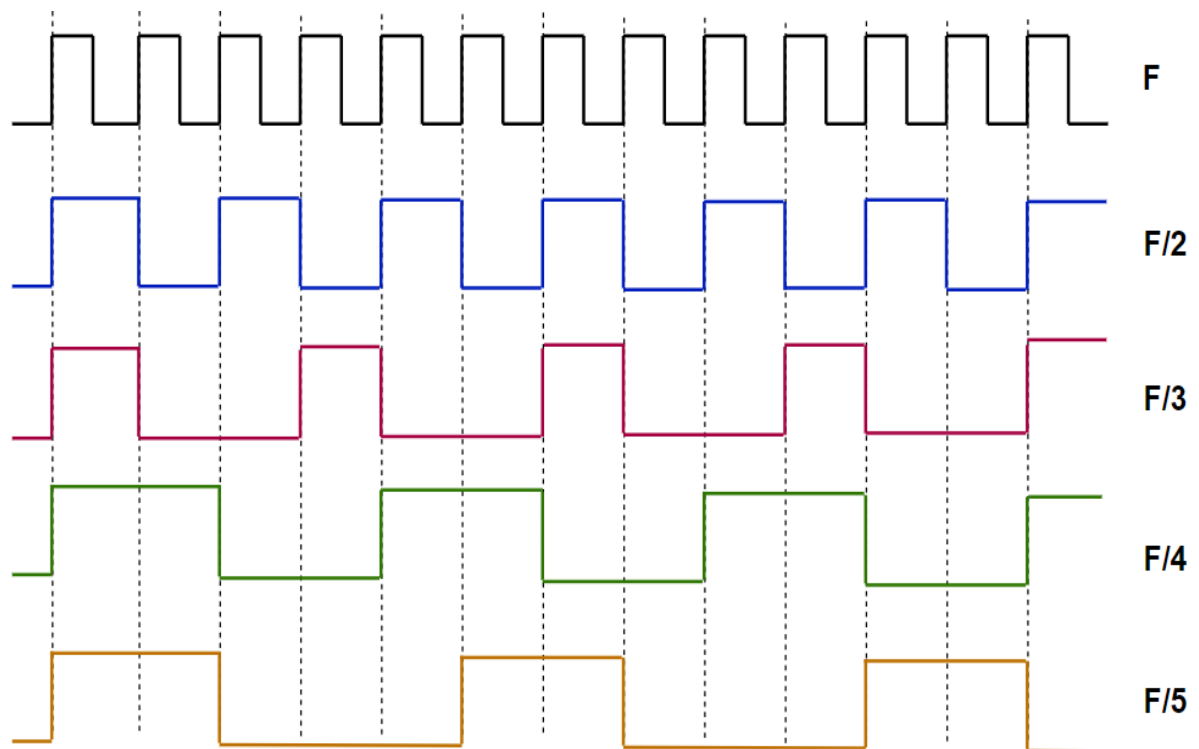
Block Interface



Ports Description

Signal Name	Description	Width
i_ref_clk	Reference Frequency	1
i_rst_n	Active Low Asynchronous Reset	1
i_clk_en	Clock Divider Block Enable	1
i_div_ratio	The divided ratio (integer value)	8

Waveforms



Corner Cases: -

1. You have to check `I_div_ratio` not equals Zero or One before enable the clock divider

```
CLK_DIV_EN = I_clk_en && ( I_div_ratio != Zero) && ( I_div_ratio != One)
```

Requirements: -

1. Write a Verilog Code to capture the above specifications
2. Write a testbench to test generation of different frequencies of both odd and even divided ratio of the reference frequency